

Man-Ki Moon

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EDUCATION

Mar. 2025 - Present	Gachon University <i>M.S. in Materials Science and Engineering, expected Feb. 2027</i> GPA: 4.31 / 4.50	Seongnam-si, Korea
Mar. 2020 - Feb. 2025	Gachon University <i>B.S. in Materials Science and Engineering</i> <i>B.S. in Industrial and Business Engineering</i> GPA: 3.83 / 4.50	Seongnam-si, Korea

RESEARCH INTERESTS

- Thin-film growth, surface chemistry, and defect control in electronic materials
- Semiconductor interfaces, contacts, dielectric integration, and low-dimensional material systems
- Thin-film process-interface relationships and data-assisted correlation of growth, structure, and device behavior

SKILLS AND TECHNIQUES

Growth and fabrication: E-beam evaporation; thermal CVD experiments with $\text{Al}_2\text{O}_3/\text{Fe}$ -coated Cu films; electrospinning; SWNT dispersion and filtration; graphene oxide (GO) and MoS_2 processing; polymer-supported metal-film test structures and thermoelectric device assembly

Interface and transport analysis: SEM-EDS; spatial and time-resolved in situ Raman spectroscopy; FTIR; quantitative image analysis; electrical, thermal, and electromechanical characterization

Computational support: Python spectral and image analysis; ORCA finite-cluster calculations; simplified OpenFOAM reactive-transport workflow

RESEARCH EXPERIENCE

Nano Synergy Laboratory, Gachon University | Graduate Researcher, Mar. 2025 - Present;
Undergraduate Researcher, Feb. 2022 - Feb. 2025

Spatial Raman Mapping, In Situ MoS_2 Vibrational Tracking, and Interfacial Contact Energetics in Electrospun PVA/PEDOT:PSS- GO/MoS_2 Thermoelectric Mats

- Fabricated electrospun PVA/PEDOT:PSS- GO/MoS_2 mats and five-unit serial devices; combined spatial Raman mapping, in situ MoS_2 vibrational tracking, FTIR, transport measurements, and ORCA interaction-energy calculations to examine interfacial reorganization.

Metal/SWNT/PI Flexible Thin-Film Conductors

- Investigated how SWNT-modified polymer interfaces influence metal-film continuity, crack evolution, and electrical reliability using controlled surface processing, e-beam evaporation, and resistance-strain measurements.
- Preliminary full-cycle resistance data suggest that an intermediate SWNT loading may better preserve electrical continuity than the metal-only control; strain-synchronized resistance and SEM crack analyses are ongoing.
- **Custom Electromechanical Testing Platform:** Designed and 3D-printed a tensile-testing system to enable synchronized mechanical loading and electrical-resistance measurements unavailable with the existing laboratory setup.

Thermal CVD Growth and Reactive-Transport Modeling of CNTs

- Deposited Al_2O_3 and Fe layers on Cu films and performed thermal CVD experiments targeting vertical CNT growth; observed CNT growth by cross-sectional SEM and developed a baseline OpenFOAM workflow coupling C_2H_2 transport with simplified surface-consumption and catalyst-response terms.

MANUSCRIPT IN PROGRESS

1. **Moon, M.-K.**, Jung, S.-C., Jang, W.-J., Beom, B.-J., and Han, J.-H. "Spatial Raman Mapping, In Situ MoS₂ Vibrational Tracking, and Interfacial Contact Energetics in Electrospun PVA/PEDOT:PSS–GO/MoS₂ Thermoelectric Mats." Manuscript in progress.

PEER-REVIEWED PUBLICATIONS

1. Beom, B.; **Moon, M.-K.**; Jung, J.-H.; Jung, S.-C.; Cho, E.-S.; Chang, K.-A.; Han, J.-H. "Polymer-Engineered MXene Composites for Durable Electrochemical Energy Storage: Suppressing Oxidation, Preserving Structure, and Extending Cycle Life." *Polymers* 18, 1365 (2026). doi:10.3390/polym18111365 Co-first author.
2. Beom, B.; Jung, S.-C.; Jang, W.; Won, J.-K.; Jeong, J.; Choi, Y.-J.; **Moon, M.-K.**; Han, J.-H. "In Situ Raman Spectroscopy-Enabled Microfluidic Gel Chromatography for Revealing Real-Time Separation Dynamics of Single-Walled Carbon Nanotubes." *Polymers* 17, 93 (2025). doi:10.3390/polym17010093
3. Jung, S.-C.; Jang, W.; Beom, B.; Won, J.-K.; Jeong, J.; Choi, Y.-J.; **Moon, M.-K.**; Cho, E.-S.; Chang, K.-A.; Han, J.-H. "Synthesis of Highly Porous Graphene Oxide-PEI Foams for Enhanced Sound Absorption in High-Frequency Regime." *Polymers* 16, 2983 (2024). doi:10.3390/polym16212983

CONFERENCE PRESENTATIONS

1. **Moon, M.-K.**, et al. "Fabrication of GO/MoS₂ Nanofiber by Electrospinning for Thermoelectric Applications." Nano Korea, 2025. Poster.

AWARDS AND HONORS

- Grand Prize, Gachon University Career Exploration Performance Report Meeting, 2023.
- Career Catch Project Scholarships, Gachon University, four awards, 2022-2024.